

Chenliang Xu

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APPOINTMENTS	Associate Professor Department of Computer Science University of Rochester, Rochester, NY	7/2022 - PRESENT
	Affiliated Faculty Goergen Institute for Data Science and Artificial Intelligence University of Rochester, Rochester, NY	9/2016 - PRESENT
	Assistant Professor Department of Computer Science University of Rochester, Rochester, NY	9/2016 - 6/2022
EDUCATION	Doctor of Philosophy Computer Science and Engineering University of Michigan, Ann Arbor, MI	2016
	Master of Science Computer Science and Engineering University at Buffalo, Buffalo, NY	2012
	Bachelor of Science Information and Computing Science Nanjing University of Aeronautics and Astronautics, Nanjing, China	2010
RESEARCH INTERESTS	My research advances multimodal AI that integrates vision, sound, and language to understand and reason about dynamic scenes. I develop methods for video understanding, audio-visual learning and generation, and multimodal reasoning, with an emphasis on making these systems reliable, interpretable, and robust.	
DISTINCTIONS	Best Demo Award Runner-Up, AAAI <i>for the demo "Caption Anything in Video"</i>	2026
	Edmund A. Hajim Outstanding Faculty Award <i>Hajim School of Engineering and Applied Sciences, University of Rochester recognizes excellence in teaching, research, and service</i>	2025
	Best Paper Honorable Mention, ACCV <i>for the paper "High-Quality Visually-Guided Sound Separation from Diverse Categories"</i>	2024
	Cisco Faculty Research Award <i>for the research on machine unlearning for trustworthy multi-modal foundation models</i>	2024
	Outstanding Speaker Award, The Physical Retail AI Workshop in WACV <i>for the invited talk on Audio-Visual Scene Synthesis for AI-Empowered Physical Retail</i>	2024
	James P. Wilmot Distinguished Professorship <i>University of Rochester recognizes promising young men and women in the early stages of their academic careers</i>	2021–2023
	University of Rochester Research Award <i>for the project "Enhancing Video Accessibility for Blind and Low-Vision Individuals Using LLMs"</i> <i>for the project "Applying AI-powered CV Techniques on Synthetic X-ray Diffraction Data"</i>	2025, 2021
	Best Paper Award, ACM SIGGRAPH VRCAI <i>for the paper "3D Human Avatar Digitization from A Single Image"</i>	2019

Tencent Faculty Research Award <i>for the research on instructional video grounding</i>	2018
Best Paper Award, The Sound and Music Computing Conference <i>for the paper "Audio-Visual Source Association for String Ensembles"</i>	2017
University of Rochester AR/VR Pilot Award <i>for the project "Real-time Synthesis of A Virtual Talking Face from Acoustic Speech"</i>	2017
Best Open Source Code Third Prize, CVPR <i>for the paper "Evaluation of Super-voxel Methods for Early Video Processing"</i>	2012
Best Demo Prize, The Greater New York Multimedia and Vision Meeting <i>for the demo "Streaming Hierarchical Video Segmentation"</i>	2012

FUNDING

Funding amounts shown beside project titles represent funds obtained as PI or my share as Co-PI or Sub-PI. Consortium totals, where provided, are identified separately.

Current Projects: *totaling \$1,903,906*
 Completed Projects: *totaling \$4,050,437*
 Internal Projects: *totaling \$127,000*
 Total Awarded Project Funding: *\$6,081,343*

Research Gifts: *totaling \$460,000*
 Pending Projects: *totaling \$650,000*

Current Projects

4. **PI: Leverage Large Language Models for Complex Robot Manipulation** **\$60,000**
 SOURCE: Center of Excellence in Data Science and AI (COE) 09/2025-12/2026
 OBJECTIVE: Develop vision-language-action models for efficient, transparent, and complex robot manipulation. Industry partner: Corning
3. **Co-PI: VIGIL–Vectors Of Intelligent Guidance In Long-Reach Rural Healthcare** **\$982,906**
 SOURCE: The Advanced Research Projects Agency for Health (ARPA-H) 11/2024-10/2029
 OBJECTIVE: Improve rural healthcare responses through a continually-learning augmented intelligence that adaptively transforms generalists into specialists based on experience, skill, and context. Responsible for multimodal modeling in a University of Michigan-led team spanning eight universities and RTX BBN; consortium funding up to \$25 million
2. **Co-PI: Artificial Intelligence for effective communication to promote vaping cessation on social media** **\$376,000**
 SOURCE: National Institutes of Health (NIH) 9/2024-8/2029
 OBJECTIVE: Provide methods effective in delivering the risks of e-cigarette use to social media users and the public, and reaching communities on the potential harms of e-cigarette use
1. **PI: Multimodal Video Understanding** **\$485,000**
 SOURCE: Sony Group Corporation (SONY) 10/2022-8/2027
 OBJECTIVE: Develop video understanding methods that interact efficiently across vision, audio, and language modalities, with rich user interaction and spatiotemporal awareness capabilities

Completed Projects

12. **Co-PI: CDS&E: Inferring Lattice Dynamics from Temporal X-ray Diffraction Data** **\$155,557**
 SOURCE: National Science Foundation (NSF) 8/2022-7/2026
 OBJECTIVE: Develop machine learning methods to detect rare lattice-level mechanisms responsible for phase transformation and plastic deformation under extreme conditions
11. **Co-PI: Time-resolved classification of x-ray diffraction data using deep-learning-powered computer vision techniques** **\$275,741**
 SOURCE: National Nuclear Security Administration (NNSA) 7/2022-6/2026
 OBJECTIVE: Develop automated deep-learning-powered computer vision techniques for classifying crystal structures and phase fractions from synthetic X-ray diffraction data

10. **Co-PI: MILLY–Multi-directional Loosely-Linked Archetype Models for Perceptually-enabled Task Guidance** **\$959,641**
SOURCE: Defense Advanced Research Projects Agency (DARPA) 1/2022-12/2024
OBJECTIVE: Develop AR/VR technologies that allow the AI assistant to perceive the user's intention and environment and provide them with just-in-time feedback through speech and aligned graphics. Group contribution: egocentric audio-visual understanding and task-procedure planning

9. **Sub-PI: From Human-Powered to Automated Video Description for Blind and Low Vision Users** **\$493,380**
SOURCE: National Institutes of Health (NIH) 7/2023-10/2024
OBJECTIVE: Develop an AI-based tool in collaboration with sighted describers and blind and low-vision individuals that more efficiently produces video descriptions and offers on-demand access to visual information in online videos

8. **PI: Towards Robust and Fair AI Algorithms against Multiple Shortcuts** **\$120,000**
SOURCE: Center of Excellence in Data Science and AI (COE) 10/2021-8/2024
OBJECTIVE: Design a novel, scalable human-machine collaborative system to assist humans in discovering unknown biases in any image classifiers. Industry partner: Kitware

7. **PI: Egocentric 3D Audio-Visual Scene Understanding** **\$399,753**
SOURCE: Meta Platforms, Inc. (Meta) 9/2021-8/2024
OBJECTIVE: Develop AI algorithms that integrate surrounding 3D auditory and visual percepts for scene understanding via audio-visual grounding, separation, and generation with egocentric videos and AR applications as foci

6. **PI: III: Small: Collaborative Research: Scalable Deep Bayesian Tensor Decomposition** **\$199,134**
SOURCE: National Science Foundation (NSF) 10/2019-9/2023
OBJECTIVE: Develop new tensor decomposition methods to discover various complicated entity relationships and handle the rapid data growth of real applications

5. **PI: BIGDATA: F: Audio-Visual Scene Understanding** **\$666,000**
SOURCE: National Science Foundation (NSF) 9/2017-8/2022
OBJECTIVE: Develop algorithms to achieve a human-like audio-visual bimodal scene understanding that overcomes the limitations in single-modality analysis through big data analysis of Internet videos

4. **Co-PI: CDS&E: D3SC: Applying Video Segmentation to Coarse-grain Mapping Operators in Molecular Simulations** **\$200,000**
SOURCE: National Science Foundation (NSF) 8/2018-7/2022
OBJECTIVE: Apply advances in computer vision, e.g., video segmentation and action recognition, to improve models of multiscale systems in chemistry

3. **PI: RI: Small: Learning Dynamics and Evolution towards Cognitive Understanding of Videos** **\$465,990**
SOURCE: National Science Foundation (NSF) 9/2018-8/2021
OBJECTIVE: Develop computational models to study higher-order inference in understanding web instructional videos.

2. **Sub-PI: BOCA – BODy-worn Camera Analysis in Public Safety** **\$74,982**
SOURCE: National Institute of Standards and Technology (NIST) 7/2019-11/2020
OBJECTIVE: Investigate contextual information (e.g., detected objects, body parts, and their spatiotemporal relations, detected audio events in the scene) to improve the first-person-view activity recognition performance.

1. **PI: Anomaly Detection for Videos Applied in Manufacturing Settings** **\$40,259**
SOURCE: Center of Excellence in Data Science and AI (COE) 10/2019-3/2020
OBJECTIVE: Develop a state-of-the-art deep learning-based computer vision framework for anomaly detection videos applied in manufacturing processes. Industry partner: Corning

Internal Projects

4. **PI: Enhancing Video Accessibility for Blind and Low-Vision Individuals Using Large Language Models** **\$42,000**
 SOURCE: University of Rochester Research Awards (UR) 7/2025-6/2026
 OBJECTIVE: A seed grant for developing an AI-powered system that enhances video accessibility and engagement for BLV individuals
3. **Co-PI: Artificial Intelligence for effective communication on health effects of electronic cigarettes through Instagram** **\$20,000**
 SOURCE: University of Rochester GIDS-AI (UR) 10/2021-10/2022
 OBJECTIVE: A seed project to identify potentially effective ways of communicating with the public about the health risks of electronic cigarette use on the most popular social media platform among youth, i.e., Instagram
2. **Co-PI: Capturing nanoscale lattice variations by applying AI-powered computer vision techniques on synthetic x-ray diffraction data** **\$35,000**
 SOURCE: University of Rochester Research Awards (UR) 7/2021-6/2022
 OBJECTIVE: A seed project to generate a planning study and preliminary XRD synthetic data that can be used for future proposals to national programs for deep learning model developments.
1. **Co-PI: Real-Time Synthesis of a Virtual Talking Face from Acoustic Speech** **\$30,000**
 SOURCE: University of Rochester AR/VR Pilot Program (UR) 8/2017-7/2018
 OBJECTIVE: Develop a real-time talking face by analyzing the correlations between auditory and visual signals to assist hearing-impaired people.

Research Gifts

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| 15. Adobe, \$10,000 | 12/2024 |
| 14. Cisco Faculty Award, \$75,000 | 9/2024 |
| 13. Adobe, \$7,000 | 12/2021 |
| 12. Corning, \$70,000 | 11/2021 |
| 11. Adobe, \$10,000 | 9/2021 |
| 10. Wilmot Professorship, \$10,000 | 7/2021 |
| 9. InnoPeak Technology, \$30,000 | 3/2021 |
| 8. Adobe, \$7,000 | 3/2021 |
| 7. Corning, \$30,000 | 3/2021 |
| 6. Adobe, \$7,500 | 11/2020 |
| 5. Adobe, \$28,500 | Fall 2019 |
| 4. Corning, \$30,000 | 10/2019 |
| 3. Viscovery, \$55,000 | 9/2018 |
| 2. Tencent AI Lab Rhino-Bird, \$60,000 | 6/2018 |
| 1. Markable AI, \$30,000 | 8/2017 |

In-Kind Support

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| • Adobe, GPU Donations | 12/2024 |
| • NVIDIA, GPU Donations | 8/2017 |

Pending Projects

2. **Co-PI: Hybrid Regression, Retrieval and Reasoning (R3) Framework for Scalable Multi-modal Geo-localization** **\$500,000**
 SOURCE: Intelligence Advanced Research Projects Activity (IARPA) 1/2027-3/2028
 OBJECTIVE: Develop a hybrid regression, retrieval, and reasoning framework for scalable multimodal geo-localization for previously unseen indoor and outdoor images, videos, audios, and multimodal queries. Currently, the proposal is in Volume II negotiation

1. **PI: ResolveAI: Vision-Language Reasoning with Grounded Evidence Chains for Intelligent Customer Support Dispute Resolution** **\$150,000**
SOURCE: Airbnb 10/2026-9/2027
OBJECTIVE: Develop a vision-language reasoning system to support customer dispute resolution with text-to-image and image-to-image grounding capabilities. Currently, the proposal is in contracting

SOFTWARE & DATA SETS

Selected collaborative research software, data sets, and community resources. GitHub stars and forks are reported as of September 18, 2026.

Community research resources

Video Understanding with Large Language Models (TCSVT 2025). Curated survey resource linking papers, code, and data sets for video-language research. **3,285 GitHub stars; 151 forks.**

Video-LMM Post-Training (2025–present). Community resource on post-training and reasoning in video large multimodal models. **303 stars; 15 forks.**

Generative AI for Cel-Animation (ICCV Workshops 2025). Survey and resource collection covering animation generation methods, projects, and data sets. **214 stars.**

Research software

Zooming Slow-Mo (CVPR 2020). Joint video frame interpolation and super-resolution; released training/testing code and pretrained models. **930 stars; 163 forks.**

TDAN (CVPR 2020). Temporally deformable alignment for video super-resolution. **418 stars; 63 forks.** The method is also implemented in OpenMMLab’s MMagic toolbox.

Audio-driven talking-head generation. **ATVGnet** (CVPR 2019; **259 stars**) and **Talking-head Generation with Rhythmic Head Motion** (ECCV 2020; **209 stars**). Public implementations and model resources for speech-driven facial animation.

Avatar reconstruction and generation. **AdaSR-TalkingHead** (ICASSP 2024; **181 stars**) and **Tri²-plane** (ECCV 2024; **141 stars**). Released implementations for higher-resolution talking heads and volumetric avatar reconstruction.

Generative sound separation. **ZeroSep** (NeurIPS 2025; **64 stars**) and **DAVIS** (ACCV 2024; **36 stars**). Public implementations for separation using pretrained diffusion models and visually guided generative modeling.

LIBSVX (2012–2016). Supervoxel segmentation library and quantitative benchmark for early video processing. **CVPR 2012 Best Open Source Code Third Prize.**

Data sets and benchmarks

TDMM-LM facial animation data (CVPR 2026). Text-paired facial videos with 3D parameter annotations; the repository provides Part 1 (approximately **70 hours**) of the approximately 80-hour corpus. **455 stars.**

OPENXRD (Digital Discovery 2026). **217 expert-curated crystallography questions**, supporting reference passages, and evaluation code for testing language and multimodal models on X-ray diffraction reasoning.

VidComposition (CVPR 2025). **982 videos and 1,706 multiple-choice questions** for evaluating video composition understanding, including cinematography, narrative, and character interpretation.

RWAVS / AV-NeRF (NeurIPS 2023). Released real-world audio-visual scene data covering **13 scenes**, with spatial audio, video, camera information, and training/evaluation code for novel-view synthesis.

UrbanCars and ImageNet-W (CVPR 2023). Data-generation/evaluation resources and model code for studying multiple spurious correlations and robustness to visual shortcuts.

MUSIC-AVQA (CVPR 2022). **45,867 question-answer pairs across 9,288 videos**, covering over 150 hours and 22 instruments; released data and code for audio-visual question answering.

Look, Listen, and Parse (LLP) (ECCV 2020). **11,849 ten-second clips across 25 event categories**, with audio/visual event annotations, features, and baseline code. Used in subsequent audio-visual parsing research, including UWAV (CVPR 2025).

Grounded Image Editing Request (GIER) (ACCV 2020). **6,179 image pairs**, averaging 4.83 language requests per pair, with editing-operation annotations and local-operation masks.

Audio-Visual Event (AVE) (ECCV 2018). **4,143 videos across 28 event categories**, temporally annotated for audio-visual event localization; released data, features, and baseline code. **211 stars**.

YouCook2 (AAAI 2018). **2,000 instructional videos, 89 recipes, and 176 hours**, with temporally localized procedure steps and language descriptions.

Actor-Action Dataset (A2D) (CVPR 2015). **3,782 videos**, seven actor classes, and eight action classes plus a no-action class; pixel-level actor/action labels for sampled frames.

PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?user=54HfyDIAAAAJ>
Citations: over 12,200, h-index: 47 (September 2026)

2026

150. Mingqian Feng, Xiaodong Liu, Weiwei Yang, Chenliang Xu, Christopher White, and Jianfeng Gao. Statistical estimation of adversarial risk in large language models under best-of-n sampling. In *Neural Information Processing Systems*, pages 1–32, 2026. (NeurIPS)
149. Jing Bi, Junjia Guo, Susan Liang, Guangyu Sun, Luchuan Song, Yunlong Tang, Jinxi He, Jiarui Wu, Ali Vosoughi, Chen Chen, and Chenliang Xu. Verify: A benchmark of visual explanation and reasoning for investigating multimodal reasoning fidelity. In *Conference on Language Modeling*, pages 1–11, 2026. (COLM)
148. Jing Bi, Luchuan Song, Dingxin Zhang, Pinxin Liu, Guangyu Sun, Lianggong Bruce Wen, Weidong Cai, Chen Chen, and Chenliang Xu. Does reasoning improve seeing? understanding when vision-language models benefit from thinking. In *International Conference on Machine Learning*, pages 1–13, 2026. (ICML)
147. Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Yunlong Tang, Zeliang Zhang, Haozheng Luo, Yuwei Han, Jianshu Zhang, Dennis Wu, Hong-Yu Chen, Haoran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, and Liu Han. Advevo-marl: Shaping internalized safety through adversarial co-evolution in multi-agent reinforcement learning. In *International Conference on Machine Learning*, pages 1–13, 2026. (ICML)
146. Chao Huang, Zeliang Zhang, Jiang Liu, Ximeng Sun, Jialian Wu, Xiaodong Yu, Ze Wang, Chenliang Xu, Emad Barsoum, and Zicheng Liu. Drift: Transferring reasoning priors for efficient mllm fine-tuning. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 20294–20309, 2026. (ACL Findings)
145. Yang Lu, Jiamu Tang, Jiankun Yang, Shijian Luo, Chenliang Xu, and Yukang Yan. Tuning the face: Modulating facial expressions for realistic self-avatars in virtual reality. In *ACM Designing Interactive Systems Conference*, pages 1419–1436, 2026. (DIS)
144. Yolo Y Tang, Chao Huang, Susan Liang, Jing Bi, Yicheng Wang, Daiki Shimada, and Chenliang Xu. Asynchronous temporal modeling with two-agent framework for streaming dense video captioning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2799–2810, 2026. (CVPR)
143. Luchuan Song, Pinxin Liu, Haiyang Liu, Zhenchao Jin, Yolo Yunlong Tang, Zichong Xu, Susan Liang, Jing Bi, Jason J Corso, and Chenliang Xu. Bridging facial understanding and animation via language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17557–17567, 2026. (CVPR)
142. Jing Bi, Filippas Bellos, Junjia Guo, Yayuan Li, Chao Huang, Yunlong Tang, Luchuan Song, Susan Liang, Zhongfei Zhang, Jason J. Corso, and Chenliang Xu. When to think and when to look: Uncertainty-guided lookback. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5104–5113, 2026. (CVPR)
141. Yolo Yunlong Tang, Daiki Shimada, Hang Hua, Chao Huang, Jing Bi, Rogerio Feris, and Chenliang Xu. Video-r4: Reinforcing text-rich video reasoning with visual rumination. In *Findings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8314–8325, 2026. (CVPR Findings)

140. Jing Bi, Yunlong Tang, Pinxin Liu, Chao Huang, Ali Vosoughi, Jiarui Wu, Jinxi He, and Chenliang Xu. I2g: Generating instructional illustrations via text-conditioned diffusion. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 4749–4759, 2026. **(CVPR Workshops)**
139. Jing Bi, Yunlong Tang, Chao Huang, and Chenliang Xu. What happens next? memory-driven action planning for egocentric storytelling characters. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 4760–4769, 2026. **(CVPR Workshops)**
138. Susan Liang, Chao Huang, Filippos Bellos, Yolo Yunlong Tang, Qianxiang Shen, Jing Bi, Luchuan Song, Zeliang Zhang, Jason J. Corso, and Chenliang Xu. Omni-judge: Can omni-llms serve as human-aligned judges for text-conditioned audio-video generation? In *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 1–12, 2026. **(CVPR Workshops)**
137. Ali Vosoughi, Yongyi Zang, Qihui Yang, Nathan Paek, Randal Leistikow, and Chenliang Xu. Multimodal room impulse response generation through latent rectified flow matching. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 14627–14631, 2026. **(ICASSP)**
136. Mingqian Feng, Xiaodong Liu, Weiwei Yang, Jialin Song, Xuekai Zhu, Chenliang Xu, and Jianfeng Gao. Sema: Simple yet effective learning for multi-turn jailbreak attacks. In *International Conference on Learning Representations*, pages 1–37, 2026. **(ICLR)**
135. Zeliang Zhang, Xiaodong Liu, Hao Cheng, Hao Sun, Chenliang Xu, and Jianfeng Gao. Training large reasoning models efficiently via progressive thought encoding. In *International Conference on Learning Representations*, pages 1–15, 2026. **(ICLR)**
134. Yolo Yunlong Tang, Jing Bi, Chao Huang, Susan Liang, Daiki Shimada, Hang Hua, Yunzhong Xiao, Yizhi Song, Pinxin Liu, Mingqian Feng, Junjia Guo, Zhuo Liu, Luchuan Song, Ali Vosoughi, Jinxi He, Liu He, Zeliang Zhang, Jiebo Luo, and Chenliang Xu. Caption anything in video: Fine-grained object-centric captioning via spatiotemporal multimodal prompting. In *AAAI Conference on Artificial Intelligence*, pages 41697–41699, 2026. **Best Demo Award Runner-up (AAAI)**
133. Hang Hua, Yunlong Tang, Ziyun Zeng, Liangliang Cao, Zhengyuan Yang, Hangfeng He, Chenliang Xu, and Jiebo Luo. Mmcomposition: Revisiting the compositionality of pre-trained vision-language models. *Transactions on Machine Learning Research*, pages 1–17, 2026. **(TMLR)**
132. Chao Huang, Susan Liang, Yapeng Tian, Anurag Kumar, and Chenliang Xu. High-quality sound separation across diverse categories via visually-guided generative modeling. *International Journal of Computer Vision*, 134(104):1–21, 2026. **(IJCV)**
131. Ali Vosoughi, Ayoub Shahnazari, Yufeng Xi, Zeliang Zhang, Griffin Hess, Chenliang Xu, and Niaz Abdolrahim. Openxrd: a comprehensive benchmark framework for llm/mlm xrd question answering. *Digital Discovery*, 5(5):1991–2015, 2026. **(Digital Discovery)**
130. Ayoub Shahnazari, Zeliang Zhang, Sachith Dissanayake, Chenliang Xu, and Niaz Abdolrahim. Machine learning approaches for crystallographic classification from synthetic 2d x-ray diffraction data. *Journal of Applied Crystallography*, 59(1):206–224, 2026. **(J. Appl. Cryst.)**

2025

129. Jiani Liu, Zhiyuan Wang, Zeliang Zhang, Chao Huang, Susan Liang, Yunlong Tang, and Chenliang Xu. Harnessing the computation redundancy in vits to boost adversarial transferability. In *Neural Information Processing Systems*, pages 1–15, 2025. **(NeurIPS)**
128. Chao Huang, Yuesheng Ma, Junxuan Huang, Susan Liang, Yunlong Tang, Jing Bi, Wenqiang Liu, Nima Mesgarani, and Chenliang Xu. Zerosep: Separate anything in audio with zero training. In *Neural Information Processing Systems*, pages 1–23, 2025. **(NeurIPS)**

127. Yunlong Tang, Pinxin Liu, Zhangyun Tan, Mingqian Feng, Rui Mao, Chao Huang, Jing Bi, Yunzhong Xiao, Susan Liang, Hang Hua, Ali Vosoughi, Luchuan Song, Zeliang Zhang, and Chenliang Xu. Mmperspective: Do mllms understand perspective? a comprehensive benchmark for perspective perception, reasoning, and robustness. In *Neural Information Processing Systems Datasets and Benchmarks Track*, pages 1–15, 2025. **(NeurIPS)**
126. Susan Liang, Chao Huang, Yunlong Tang, Zeliang Zhang, and Chenliang Xu. p-avas: Can physics-integrated audio-visual modeling boost neural acoustic synthesis? In *IEEE/CVF International Conference on Computer Vision*, pages 13942–13951, 2025. **(ICCV)**
125. Pinxin Liu, Luchuan Song, Junhua Huang, Haiyang Liu, and Chenliang Xu. Gesturelsm: Latent shortcut based co-speech gesture generation with spatial-temporal modeling. In *IEEE/CVF International Conference on Computer Vision*, pages 10929–10939, 2025. **(ICCV)**
124. Yunlong Tang, Junjia Guo, Pinxin Liu, Zhiyuan Wang, Hang Hua, Jia-Xing Zhong, Yunzhong Xiao, Chao Huang, Luchuan Song, Susan Liang, Yizhi Song, Liu He, Jing Bi, Mingqian Feng, Xinyang Li, Zeliang Zhang, and Chenliang Xu. Generative ai for cel-animation: A survey. In *IEEE/CVF International Conference on Computer Vision Workshops*, pages 3837–3850, 2025. **(ICCV Workshops)**
123. Zeliang Zhang, Xiaodong Liu, Hao Cheng, Chenliang Xu, and Jianfeng Gao. Diversifying the expert knowledge for task-agnostic pruning in sparse mixture-of-experts. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 86–102, 2025. **(ACL Findings)**
122. Luchuan Song, Yang Zhou, Zhan Xu, Yi Zhou, Deepali Aneja, and Chenliang Xu. Streamme: Simplify 3d gaussian avatar within live stream. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Papers*, pages 1–10, 2025. **(SIGGRAPH)**
121. Susan Liang, Dejan Markovic, Israel D Gebru, Steven Krenn, Todd Keebler, Jacob Sandakly, Frank Yu, Samuel Hassel, Chenliang Xu, and Alexander Richard. Binauralflow: A causal and streamable approach for high-quality binaural speech synthesis with flow matching models. In *International Conference on Machine Learning*, pages 1–18, 2025. **(ICML)**
120. Yunlong Tang, Jing Bi, Siting Xu, Luchuan Song, Susan Liang, Teng Wang, Daoan Zhang, Jie An, Jingyang Lin, Rongyi Zhu, Ali Vosoughi, Chao Huang, Zeliang Zhang, Feng Zheng, Jianguo Zhang, Ping Luo, Jiebo Luo, and Chenliang Xu. Video understanding with large language models: A survey. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(2):1355–1376, 2025. **(TCSVT)**
119. Zeliang Zhang, Gaowen Liu, Charles Fleming, Ramana Rao Kompella, and Chenliang Xu. Targeted forgetting of image subgroups in clip models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9870–9880, 2025. **(CVPR)**
118. Yunlong Tang, Junjia Guo, Hang Hua, Susan Liang, Mingqian Feng, Xinyang Li, Rui Mao, Chao Huang, Jing Bi, Zeliang Zhang, Pooyan Fazli, and Chenliang Xu. Vidcomposition: Can mllms analyze compositions in compiled videos? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8490–8500, 2025. **(CVPR)**
117. Chao Huang, Ruohan Guo, J. M. F. Tsang, Jan Kurcius, Cagdas Bilen, Chenliang Xu, Anurag Kumar, and Sanjeel Parekh. Learning to highlight audio by watching movies. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23925–23935, 2025. **(CVPR)**
116. Jing Bi, Lianggong Bruce Wen, Zhang Liu, Junjia Guo, Yunlong Tang, Bingjie Wang, and Chenliang Xu. Unveilling visual perception in language models: An attention head analysis approach. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4135–4144, 2025. **(CVPR)**
115. Zeliang Zhang, Susan Liang, Daiki Shimada, and Chenliang Xu. Rethinking audio-visual adversarial vulnerability from temporal and modality perspectives. In *International Conference on Learning Representations*, pages 1–16, 2025. **(ICLR)**
114. Pinxin Liu, Luchuan Song, Daoan Zhang, Hang Hua, Yunlong Tang, Huaijin Tu, Jiebo Luo, and Chenliang Xu. Gaussianstyle: Gaussian head avatar via stylegan. In *International Conference on 3D Vision*, pages 1–10, 2025. **(3DV)**

113. Yunlong Tang, Daiki Shimada, Jing Bi, Mingqian Feng, Hang Hua, and Chenliang Xu. Empowering llms with pseudo-untrimmed videos for audio-visual temporal understanding. In *AAAI Conference on Artificial Intelligence*, pages 1–15, 2025. **(AAAI)**
112. Yunlong Tang, Gen Zhan, Li Yang, Yiting Liao, and Chenliang Xu. Cardiff: Video salient object ranking chain of thought reasoning for saliency prediction with diffusion. In *AAAI Conference on Artificial Intelligence*, pages 1–10, 2025. **(AAAI)**
111. Hang Hua, Yunlong Tang, Chenliang Xu, and Jiebo Luo. V2xum-llm: Cross-modal video summarization with temporal prompt instruction tuning. In *AAAI Conference on Artificial Intelligence*, pages 1–13, 2025. **(AAAI)**
110. Zidian Xie, Nanda Kishore Korrapolu, Amisha Dubey, Luchuan Song, Chenliang Xu, Karen M Wilson, AnaPaula Cupertino, and Dongmei Li. Leveraging large language models to identify engagement-driving features in vaping-related tiktok videos: Cross-sectional study. *Journal of Medical Internet Research*, 27:1–11, Nov 2025. **(JMIR)**
109. Matthew Xie and Chenliang Xu. Predicting the risk of asthma development in youth using machine learning models. *PloS one*, 20(11):1–16, 2025. **(PLOS One)**

2024

108. Susan Liang, Chao Huang, Yapeng Tian, Anurag Kumar, and Chenliang Xu. Language-guided audio-visual editing via one-shot adaptation. In *Asian Conference on Computer Vision*, pages 1011–1027, 2024. **(ACCV)**
107. Chao Huang, Susan Liang, Yapeng Tian, Anurag Kumar, and Chenliang Xu. High-quality visually-guided sound separation from diverse categories. In *Asian Conference on Computer Vision*, pages 35–49, 2024. **Best Paper Honorable Mention (ACCV)**
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22. Tianlang Chen, Chenliang Xu, and Jiebo Luo. Improving text-based person search by spatial matching and adaptive threshold. In *IEEE Winter Conference on Applications of Computer Vision*, pages 1879–1887, 2018. **(WACV)**
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16. Yan Yan, Chenliang Xu, Dawen Cai, and Jason J. Corso. Weakly supervised actor-action segmentation via robust multi-task ranking. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1022–1031, 2017. **(CVPR)**
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12. Luowei Zhou, Chenliang Xu, Parker Koch, and Jason J. Corso. Watch what you just said: Image captioning with text-conditional attention. In *ACM International Conference on Multimedia Thematic Workshops*, pages 305–313, 2017. **(Multimedia Workshops)**

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10. Chenliang Xu and Jason J. Corso. Libsvx: A supervoxel library and benchmark for early video processing. *International Journal of Computer Vision*, 119(3):272–290, 2016. **(IJCV)**
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5. Pradipto Das, Chenliang Xu, Richard F. Doell, and Jason J. Corso. A thousand frames in just a few words: Lingual description of videos through latent topics and sparse object stitching. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2634–2641, 2013. **(CVPR)**
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2. S. Oh, A. Perera, I. Kim, M. Pandey, K. Cannons, H. Hajimirsadeghi, A. Vahdat, G. Mori, B. Miller, S. McCloskey, Y.-C. Cheng, Z. Huang, C.-H. Lee, C. Xu, R. Kumar, W. Chen, J. J. Corso, L. Fei-Fei, D. Koller, V. Ramanathan, K. Tang, A. Joulin, and A. Alahi. Trecvid 2013 genie: Multimedia event detection and recounting. In *NIST TRECVID Workshop*, 2013. **(TRECVID)**
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**SELECTED INVITED
TALKS & WORKSHOP
KEYNOTES**

- Multi-level Alignment in Audio-Visual Generation and Learning
 - *Keynote Speaker*, ACM Multimedia Workshop on Joint Audio-Video Comprehension and Generation, November 2026 (scheduled)
 - Johns Hopkins University, October 2026 (scheduled)
 - *Keynote Speaker*, ACL MAGMaR: Workshop on Multimodal Augmented Generation via Multimodal Retrieval, July 2026
 - Institute for Pure and Applied Mathematics, May 2026
 - University of Michigan, Ann Arbor, March 2026
 - *Keynote Speaker*, ICCV Workshop on Audio-Visual Generation and Learning, October 2025
- Understand and Reconstruct Multimodal Egocentric Scenes
 - Sony, March 2025
 - VinAI, January 2025
 - Hanoi University of Science and Technology, January 2025
 - Corning, Inc., August 2024
- Audio-Visual Synthesis, Separation, and Joint Editing by Language
 - ECCV Workshop on AVGenL, September 2024
- Towards Fair Computer Vision: Discover the Hidden Biases of an Image Classifier
 - AI and Computer Vision Meetup by Voxel5, February 2024
- Audio-Visual Scene Synthesis for AI-Empowered Physical Retail
 - WACV Workshop on Physical Retail AI, January 2024
- The Fragile Audio-Visual Integration under Adversarial Attacks
 - CVPR Workshop on Sight and Sound, June 2021
- Cyclic Co-Learning of Sounding Object Visual Grounding and Sound Separation
 - Facebook Reality Labs, April 2021
- Discover the Hidden Biases of an Image Classifier
 - URCS Computer Vision Seminar, February 2021
 - Aibee Inc., January 2021

Learning from Web Instructional Videos

- NIST PCSR PI Meeting, July 2020
- UR Warner School LiDA Colloquium Series, February 2020

Audio-Visual Scene Understanding

- MIT Computer Vision Seminar, October 2020
- Corning Inc., November 2019
- Viscovery Inc., May 2019
- UR Laboratory for Laser Energetics, December 2018
- Western New York Image and Signal Processing Workshop, October 2018
- UR Data Science Summer Colloquium, June 2018
- Rochester Institute of Technology, November 2017

Applications of Multimodal Deep Learning

- Clinical & Translational Science Institute, April 2018

Towards the What, Who, and Where of Video Understanding

- CVPR Workshop on The DAVIS Challenge on Video Object Segmentation, July 2017
- Midwest Vision Workshop, April 2016

Scale-Adaptive Video Understanding

- University of Utah, March 2016
- Stevens Institute of Technology, March 2016
- University of Rochester, March 2016
- Michigan State University, February 2016

UNIVERSITY TEACHING

CSC 249/449 Machine Vision

Spring 2018, Spring 2019, Spring 2020, Fall 2020, Spring 2022, Spring 2024, Spring 2025, Spring 2026
Redesigned the course around modern computer vision, developing instructional materials and assignments, and updating them as the field advances. Taught seven full offerings and assumed instructional responsibility for the second half of Spring 2026 to maintain course continuity during a colleague's leave.

CSC 245/445 Deep Learning

Previously CSC 298/578; renumbered in 2023.

Spring 2017, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025, Fall 2026

Developed the University of Rochester's first dedicated Deep Learning course for undergraduate and graduate students. Continually update course materials to reflect advances in deep learning and generative AI. Completed six offerings; teaching the seventh in Fall 2026.

CSC 577 Advanced Topics in Computer Vision

Fall 2016, Fall 2017, Fall 2018, Fall 2019, Spring 2021, Spring 2023

ECE 410 Introduction to Augmented and Virtual Reality (Co-Instructor)

Fall 2020, Fall 2021, Fall 2022, Fall 2023, and Fall 2024

ECE 411 Selected Topics in Augmented and Virtual Reality (Co-Instructor)

Spring 2023

INTERDISCIPLINARY RESEARCH TRAINING **NSF NRT program on Augmented and Virtual Reality**

Faculty Participant, 2019–2024

Contributed to trainee selection, co-taught five offerings of ECE 410 and one offering of ECE 411, and mentored graduate trainees in interdisciplinary research.

NSF REU Site on Computational Methods for Understanding Music, Media, and Minds

Faculty Participant, 2017–2020

Reviewed applications, taught deep learning bootcamp workshops, and mentored six undergraduate researchers.

STUDENT ADVISING Served as primary advisor to six Ph.D. graduates and currently advise five Ph.D. students as primary advisor and one as co-advisor. Graduates' first appointments include a tenure-track faculty position and research scientist positions in industry.

Completed Ph.D. Students—Primary Advisor

6. Chao Huang (UR CS) Ph.D., 2026
DISSERTATION: *Controllable Generative Modeling for Multimodal Perception and Synthesis*
FIRST APPOINTMENT: *Research Scientist at Tencent Hunyuan*

5. Jing Bi (UR CS) Ph.D., 2026
DISSERTATION: *Learn to Act in Pixels*
FIRST APPOINTMENT: *Research Scientist at ByteDance*
4. Zhiheng Li (UR CS) Ph.D., 2023
DISSERTATION: *Discover and Mitigate Biases in Discriminative and Generative Image Models*
FIRST APPOINTMENT: *Research Scientist at Amazon AWS*
CURRENT POSITION: *Member of Technical Staff on superintelligence team at Microsoft AI*
3. Jing Shi (UR CS) Ph.D., 2022
DISSERTATION: *Vision Based Language to Action Mapping*
FIRST APPOINTMENT: *Research Scientist at Adobe Research*
2. Yapeng Tian (UR CS) Ph.D., 2022
DISSERTATION: *Audio-Visual Scene Understanding Towards Unified, Explainable, and Robust Multi-sensory Perception*
FIRST APPOINTMENT: *Assistant Professor at UT Dallas*
1. Lele Chen (UR CS) Ph.D., 2022
DISSERTATION: *High-Fidelity Talking Avatar Video Generation*
FIRST APPOINTMENT: *Staff Research Scientist at Oppo US Research*
CURRENT POSITION: *Research Scientist at Meta Superintelligence Lab*

Current Ph.D. Students

Primary advisor unless otherwise indicated.

6. Mingqian Feng (UR CS) 2023—Expected 2028
5. Yunlong Tang (UR CS) 2023—Expected 2027
4. Susan Liang (UR CS) 2022—Expected 2027
3. Zeliang Zhang (UR CS) 2022—Expected 2027
2. Luchuan Song (UR CS) 2022—Expected 2026
1. Ali Vosoughi (UR ECE, co-advising) 2022—Expected 2026

Student and Alumni Distinctions

Selected individual and shared awards; alumni achievements after graduation are identified below.

Jiani Liu, Undergraduate Research Advisee—Schwartz Discover Grant, 2025; Runner-up, CRA Outstanding Undergraduate Researcher Awards, 2026.

Zhangyun Tan, Undergraduate Research Advisee—CS Discover Grant, University of Rochester, 2025.

Chao Huang, Ph.D. Advisee (2026)—NeurIPS Scholar Award, 2025; ICCV Doctoral Consortium, 2025; ICLR Top 200 Reviewers, 2026.

Susan Liang, Ph.D. Advisee—ICML Scholar Award, 2025.

Yunlong Tang, Ph.D. Advisee—First Place (team award), AIM Challenge on Video Saliency Prediction, ECCV Workshops, 2024.

Zhiheng Li, Ph.D. Advisee (2023)—NeurIPS Top Reviewer, 2022; CVPR Doctoral Consortium, 2023.

Jing Shi, Ph.D. Graduate (2022)—ICML Golden Reviewer, 2026 (post-graduation).

Yapeng Tian, Ph.D. Alumnus (2022)—CVPR Doctoral Consortium, 2022; post-graduation distinctions: AAAI New Faculty Highlights, 2023; Cisco Faculty Research Award, 2023; Amazon Research Award, 2024; UT Dallas ECS Faculty Teaching Award, 2026.

Sizhe (Lester) Li, Undergraduate Research Alumnus—Finalist, CRA Outstanding Undergraduate Researcher Awards, 2023; MIT Presidential Fellowship; coauthor, *pixelSplat*, CVPR Best Paper Runner-Up, 2024 (post-Rochester).

Lele Chen, Ph.D. Advisee (2022)—Coauthor, *3D Human Avatar Digitization from a Single Image*, VRCAI Best Paper Award, 2019; Donald M. and Janet C. Barnard Fellowship, 2020.

Matthew Xie, High School Research Advisee—Top-300 Scholar, Regeneron Science Talent Search, 2025.

Shared paper and demonstration awards: Chao Huang, Susan Liang, and alumnus Yapeng Tian were coauthors of *DAVIS*, ACCV Best Paper Honorable Mention, 2024. Yunlong Tang and student/alumni coauthors received the AAAI Best Demonstration Award Runner-up for *Caption Anything in Video*, 2026.

Master's Research Advisees

22. Junjia Guo (Summer 2024 - Summer 2025, then a Ph.D. student at the University of Arizona)
21. Bowen Yang (Spring 2024 - Summer 2025)
20. Nguyen Manh Nguyen (Fall 2022 - Spring 2024, then Senior Applied Scientist at Aitomatic)
19. Rongyi Zhu (Fall 2022 - Spring 2024, then a Ph.D. student at Stony Brook University)
18. Shijian Deng (Fall 2021 - Fall 2022, then a Ph.D. student at UT Dallas)
17. Guangyu Sun (Fall 2020 - Summer 2022, then a Ph.D. student at UCF)
16. Rohan Sharma (Summer - Fall 2020, then a Ph.D. student at SUNY Buffalo)
15. Guo Li (Summer 2019 - Spring 2020, then Amazon)
14. Guofeng Cui (Spring 2019 - Spring 2020, then a Ph.D. student at Rutgers University)
13. Ziyi Kou (Fall 2018 - Spring 2020, then a Ph.D. student at Univ. of Notre Dame)
12. Justin Tian (Fall 2018 - Spring 2019)
11. Jing Bi (Summer 2018 - Summer 2020, then a Ph.D. student in URCS)
10. Wei Zhang (Summer 2018 - Spring 2019, then Amazon)
9. Hao Huang (Summer 2018 - Spring 2019, then a Ph.D. student at NYU)
8. Wentian Zhao (Summer 2018 - Spring 2019, then Adobe Research)
7. Shaojie Wang (Summer 2018 - Spring 2019, then a Ph.D. student at WUSTL)
6. Jie Chen (Fall 2017 - Fall 2019, then Microsoft Research)
5. Li Ding (Summer 2017, then a Research Engineer at MIT)
4. Sudhanshu Srivastava (Spring - Summer 2017, then a Ph.D. student at UCSB)
3. Lele Chen (Spring 2017 - Spring 2018, then a Ph.D. student in URCS)
2. Mingyang Zhou (Spring 2016, Univ. of Michigan, then a Ph.D. student at UC Davis)
1. Yao Li (Spring 2013, SUNY Buffalo)

Undergraduate Research Advisees

35. Zhangyun Tan (Spring 2025 - Present)
34. Jiani Liu (Fall 2024 - Present)
Runner-Up for the CRA Outstanding UG Researcher Awards 2026
33. Junhua Huang (Fall 2024 - Summer 2025, then a master's student at UCLA)
32. Jiarui Wu (Summer 2024 - Summer 2025, then a master's student at CMU)
31. Jinxi He (Summer 2024 - Summer 2025, then a master's student at CMU)
30. Xin Liang (Summer 2024 - Summer 2025, Tongji University)
29. Pinxin (Andy) Liu (Spring 2023 - Summer 2024, then a Ph.D. student in URCS)
28. Ji-Ze Jang (Summer 2021 - Summer 2022)
27. Miranda Rublaitus (Summer 2022, NSF REU, Yale University)
26. Kalsey Colotl (Summer 2022, NSF REU, NYU)
25. Sizhe Li (Spring 2020 - Fall 2022, then a Ph.D. student in MIT)
Finalist for the CRA Outstanding UG Researcher Awards 2023
24. Phuong Vu (Fall 2021)
23. Yiyang Su (Summer 2020 - Spring 2021, then a Ph.D. student at MSU)
22. Logan Peters (Summer 2020, Carleton College)
21. Theodore Chapman (Summer 2020, NSF REU)
20. Patrick Phillips (Summer 2020, NSF REU)
19. Bryce Elizabeth Yahn (Summer 2020, NSF REU)

18. Tianyou Xiao (Summer 2018 - Fall 2019, then a master's student at Cornell Tech)
17. Wenxuan Bao (Summer - Fall 2019, Tsinghua University, then a Ph.D. student at UIUC)
16. Ariel Tello (Summer 2019, Xerox Research Fellow)
15. Chenxiao Guan (Summer 2018, Xerox Research Fellow, then a master's student at CMU)
14. Qiuyue Sun (Summer 2018, then a master's student at Stanford)
13. Justin Goodman (Summer 2018, NSF REU, University of Maryland)
12. Marc Moore (Summer 2018, NSF REU, Mississippi State University)
11. Amos Newswanger (Summer - Fall 2017)
10. Max Torop (Summer 2017, then a master's student at WUSTL)
9. Zhiheng Li (Summer-Fall 2017, Wuhan University, then a Ph.D. student in URCS)
8. Wei Zhao (Summer 2017, USTC)
7. Wes Smith (Summer 2017, NSF REU, University of Edinburgh, then a Ph.D. student in URCS)
6. Moses Bug (Summer 2017, NSF REU, Brandeis University)
5. Shengqi Suizhu (Fall 2016 - Spring 2017)
4. Austin Schaffer (Spring 2014, SUNY Buffalo)
3. Libing Wu (Spring 2014, SUNY Buffalo)
2. Spencer Whitt (Summer 2013, SUNY Buffalo)
1. Tyler Ganter (Summer 2013, SUNY Buffalo)

Visiting Scholars

- | | |
|---|-------------------------|
| 3. Daiki Shimada
<i>Researcher at Sony R&D Center</i> | Fall 2022—Summer 2023 |
| 2. Robert Jacobs
<i>Professor of Brain and Cognitive Science, University of Rochester</i> | Fall 2018 |
| 1. Baojie Fan
<i>Associate Professor, Nanjing University of Posts and Telecommunications</i> | Summer 2018—Spring 2019 |

Doctoral Dissertation Committees

- | | |
|--------------------------------------|----------------|
| 39. Yongsheng Yu (UR CS) | |
| 38. Jinfa Huang (UR CS) | |
| 37. Ayoub Shahnazari (UR ME) | |
| 36. Jingyang Lin (UR CS) | |
| 35. Yifan Li (UR CS) | |
| 34. Yifan Zhu (UR CS) | |
| 33. James Spann (UR CS) | |
| 32. Hang Hua (UR CS) | Graduated 2025 |
| 31. Nathan Kent (UR CS) | Graduated 2025 |
| 30. Jie An (UR CS) | Graduated 2025 |
| 29. Max Wasserman (UR CS) | Graduated 2025 |
| 28. Ruoyang Hu (UR BCS/CS) | Graduated 2024 |
| 27. Yiming Gan (UR CS) | Graduated 2023 |
| 26. Yu Feng (UR CS) | Graduated 2023 |
| 25. Yu Chen (UR CS) | Graduated 2023 |
| 24. Songyang Zhang (UR CS) | Graduated 2023 |
| 23. Haitian Zheng (UR CS) | Graduated 2023 |
| 22. Wei Zhu (UR CS) | Graduated 2023 |
| 21. Ali K. Shargh (UR ME) | Graduated 2022 |
| 20. Sabyasachi Shivkumar (UR BCS/CS) | Graduated 2022 |

19. Weijian Li (UR CS)	Graduated 2022
18. Wei Xiong (UR CS)	Graduated 2022
17. Joseph German (UR BCS/CS)	Graduated 2022
16. Lisa Jin (UR CS)	Graduated 2022
15. Zhengyuan Yang (UR CS)	Graduated 2021
14. Shupeng Gui (UR CS)	Graduated 2021
13. Hanlin Tang (UR CS)	Graduated 2021
12. Samiha Samrose (UR CS)	Graduated 2021
11. Taylan Sen (UR CS)	Graduated 2021
10. Nabil Hossain (UR CS)	Graduated 2020
9. Haichuan Yang (UR CS)	Graduated 2020
8. Feng Yang (UR CS)	Graduated 2020
7. Christopher Bates (UR BCS/CS)	Graduated 2020
6. Bochen Li (UR ECE)	Graduated 2020
5. Kushal Kafle (Rochester Institute of Technology)	Graduated 2020
4. Haofu Liao (UR CS)	Graduated 2019
3. Jianbo Yuan (UR CS)	Graduated 2019
2. Yichi Zhang (UR ECE)	Graduated 2019
1. Sefik Emre Eskimez (UR ECE)	Graduated 2019

PROFESSIONAL SERVICE

Leadership and Editorial Roles

• Program Chair, WACV	2028 (upcoming)
• Program Co-Chair, ICRAS	2026, 2027
• Area Chair, CVPR	2024—present
• Area Chair, ACM Multimedia	2023—present
• Associate Editor, IEEE Transactions on Multimedia	2022—2024
• Senior PC, AAAI	2022
• Senior PC, IJCAI	2021
• Advisory Committee, CVIP	2021

Workshop and Tutorial Organization

• Organizer, CVPR Workshop on GenAI for Storytelling	2026
• Organizer, AVSS Tutorial on Multimedia Deepfake Detection	2025
• Organizer, ICCV Workshop on Trustworthy Foundation Models	2025
• Organizer, ICME Tutorial on Multimedia Deepfake Detection	2024
• Organizer, CVPR Tutorial on Audio-Visual Scene Understanding	2021
• Organizer, WACV Tutorial on Audio-Visual Scene Understanding	2021
• Organizer, CVPR Workshop on Motion and Spatio-Temporal Representations	2017
• Presenter, IEEE CVPR Tutorial on Video Segmentation	2014

Conference Reviewer

• IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	2016—2023
• European Conference on Computer Vision (ECCV)	2018—2022
• IEEE/CVF International Conference on Computer Vision (ICCV)	2015—2025
• Neural Information Processing Systems (NeurIPS)	2016, 2021, 2025
• International Conference on Learning Representations (ICLR)	2022, 2025
• AAAI Conference on Artificial Intelligence (AAAI)	2017—2021
• Conference on Empirical Methods in Natural Language Processing (EMNLP)	2020
• IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)	2026
• Asian Conference on Computer Vision (ACCV)	2016—2018
• British Machine Vision Conference (BMVC)	2015—2019, 2021
• Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)	2019
• Annual Conference of the European Association for Computer Graphics	2019

- International Conference on Semantic Computing (ICSC) 2017
- International Symposium CompIMAGE'16 2016
- XXIIIrd ISPRS Congress 2016
- ISPRS Geospatial Week – Image Sequence Analysis 2015
- IEEE/ISPRS 3rd Joint Workshop on Multi-Sensor Fusion for Dynamic Scene Understanding 2015
- Indian Conference on Computer Vision, Graphics and Image Processing 2014

Journal Reviewer

- Reviewer, IEEE Transactions on Pattern Analysis and Machine Intelligence 2016—2021
- Reviewer, IEEE Transactions on Image Processing 2015—2021
- Reviewer, IEEE Transactions on Multimedia 2013—2021
- Reviewer, International Journal of Computer Vision 2016—2017, 2020
- Reviewer, Computer Vision and Image Understanding 2016—2020
- Reviewer, IEEE Transactions on Circuits and Systems for Video Technology 2014—2019
- Reviewer, Pattern Recognition 2016—2019
- Reviewer, Autonomous Robots 2018
- Reviewer, Signal Processing: Image Communication 2016—2017
- Reviewer, Journal of Visual Communication and Image Representation 2017
- Reviewer, ISPRS Journal of Photogrammetry and Remote Sensing 2017
- Reviewer, Machine Vision and Applications 2016
- Reviewer, IET Computer Vision 2014, 2016
- Reviewer, Electronic Letters on Computer Vision and Image Analysis 2016
- Reviewer, Image and Vision Computing 2014—2015
- Reviewer, OSA Biomedical Optics Express 2014
- Reviewer, IPSJ Transactions on Computer Vision and Applications 2013

Grant Panelist/Reviewer

- Hong Kong Research Grants Council (External Reviewer) 2026
- Austrian Science Fund (External Reviewer) 2026
- National Science Foundation (CISE/IIS, Panelist) 2021
- National Science Foundation (Ad-hoc Reviewer) 2020
- National Science Foundation (CISE/CNS, Panelist) 2019
- National Science Foundation (CISE/IIS, Panelist) 2019
- University Research Awards (UR, Reviewer) 2018

UNIVERSITY SERVICE

- UR CS Graduate Education Committee 2026—2027
- UR CS PhD Admissions Committee Co-Chair/Chair 2022—2026
Processed the largest applicant pool in the department's history, with more than 1,300 applicants; recruited the largest incoming Ph.D. cohort, with 19 students accepting offers.
- UR Data Science PhD Program Planning Committee 2022—2023
- UR CS Undergraduate Curriculum Committee 2019—2021
- UR CS Lab Committee 2018—2022
- UR CS PhD Admissions Committee 2017—2022
- UR GIDS Working Group Co-Chair 2020—2021
- UR Sproull Fellowship Review Panel 2021
- UR GIDS MS Admissions Committee 2018—2020
- UR Outstanding Dissertation Award Committee 2019
- UR CS Faculty Search Committee 2019
- UR CS Website Committee 2018

SELECTED MEDIA COVERAGE & EXPERT COMMENTARY

- “How AI Helps World Cup Referees Make the Call,” *University of Rochester News Center*, June 2026.
- Also featured by *EurekAlert!* and *Tech Xplore*.
- “Are Video Deepfakes Powerful Enough to Influence Political Discourse?,” *University of Rochester News Center*, October 2024. Also featured by *EurekAlert!*.
- “AI Analysis May Improve Vaping Cessation Efforts on Social Media,” *Clinical & Translational Science Institute Stories Blog*, September 2024.
- “Machine Learning Boosts Search for New Materials,” *University of Rochester News Center*, 2023.

"Machine Learning Pinpoints When Matter Changes under Extreme Conditions," *University of Rochester News Center*, 2022.